

SURF-DUNE interface

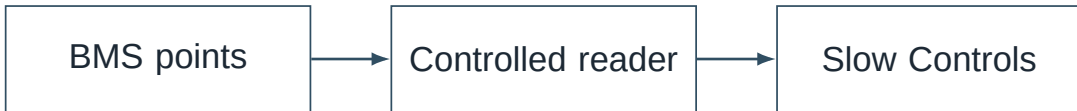
Proposed functional architecture

SURF / SOURCE

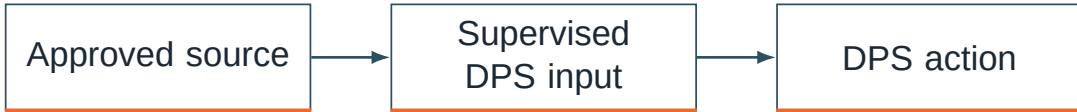
CONTROLLED BOUNDARY

DUNE

MONITORING



DETECTOR PROTECTION



The protection path is intended to work without BMS monitoring.

BMS: building management system. DPS: detector protection system.

Alarm and action ownership

FUNCTION	AUTHORITY
Facility alarm	Facility or source owner: response and reset.
DUNE alarm mirror	DUNE operator: acknowledgment of the mirror.
Data acquisition	Run Control: run state and orderly stop.
Detector protection	DPS owner: permit, inhibit, trip and reset.

DUNE acknowledgment leaves source alarms and protection latches with their owners.

Facility conditions

CONDITION	PROPOSED RESPONSE
Power / reserve	Run Control initiates the agreed orderly stop.
Network loss	Old data: STALE. Missing state: UNKNOWN.
Fire / low oxygen	Emergency response; mapped DPS action.
Liquid argon leak	Response follows an explicit source leak state.
Cooling / water	Local protection follows the agreed mapping.
Weather / access	SURF restrictions inform operations.

Protocol choices

INTERFACE	CANDIDATE	INFORMATION FLOW
BMS readout	BACnet/IP	Reader queries selected source points.
DUNE view	OPC UA	Reader publishes to Slow Controls.
Health	SNMPv3	Supported power and network telemetry.

BACnet/SC depends on SURF support.

The DPS protection interface is defined separately.

From concept to agreed interface

1 Point meaning and ownership

Each point has a defined source, state and owner.

2 Data path and timing

The interface defines read access, update rate and data age.

3 Response and recovery

Checks cover loss, approved protection action and reset.
